Anaconda v4.3 - Randomized 3D Over/Under Head Crossing

STAGE 2 IMPLEMENTATION REPORT | base file: anaconda-wordpress-snippet-v4.2.2.html (immutable start point) | deliverable: anaconda-wordpress-snippet-v4.3.html

1. Diagnosis: why every crossing read as UNDER in v4.2.2

The complete v4.2.2 pipeline was traced before any change. The rig is a trail-following chain: the head position is integrated from gait and steering, and all 79 body rings simply follow the path the head traced (**step()** chain constraint). Every ring centre is **ground-locked**: y = 0.02 + alert-lift + bob, where the lift term is non-zero only for the front ~13 rings during the alert pose. There was no vertical degree of freedom anywhere in the crossing path. The two collision systems are horizontal only: **_relaxBody()** pushes overlapping rings apart in the ground plane (XZ) and deliberately excludes head rings 0-3 (wrapped by the rigid skull loft), and **_selfAvoid()** steers the heading away from probe hits, also in the plane. So when a U-turn, a panic run or a wander target put a far body loop across the head's path, the head met that loop at **identical height**; nothing could lift either section, the v4.1 crease shading marked the overlap, and from the fixed high camera the loop visually covered the head. Every crossing therefore read as the head sliding UNDER - not because a mechanic chose UNDER, but because no over/under mechanic existed at all. Same-level intersection is deterministic; that is the root cause.

Consequence for the design: the fix cannot be a render-order trick (explicitly forbidden by the brief). The head must gain a real, physics-consistent vertical separation that is zero in free crawl and appears only when the head actually drives across its own body.

2. The randomized crossing-selection mechanism

A new per-rig system, **_updateCrossing()**, runs once per frame inside step() (before relax, contact shading and the skin rebuild, so every downstream system sees the separated geometry). It works in five stages:

(1) Capture. Pure geometry on the persistent spine path. A head-zone ring (0..14) captures a far section only if ALL gates pass: the animal is moving (speed > 0.05 - a rest guard sitting well below the cramped-space crawl throttle, so a committed approach never flickers out of capture); the section is at least 28 rings behind the head along the spine (rings 16-27 are the near-trailing body that permanently rides beside the head in the S-weave - the same rule the v4.0 self-avoid probes use) and outside the tail-drape zone (last 18 rings, which the relax guard also excludes: thin, soft, meant to drape); the section lies ahead of the nose plane; it sits in a forward corridor (forward offset must beat 1.2x lateral offset, so trailing loops beside the head never trigger); and the pair is inside a tangency-scaled anticipation band (0.72 x clamp(dT/0.62, 0.6, 1) before the surfaces touch, dT = 1.02 x (r_i + r_j)).

(2) Decision. The FIRST frame in which a pair is captured while the head is closing on it takes exactly one coin flip: mode = Math.random() < 0.5 ? OVER : UNDER. The decision is then LOCKED for the whole engagement - it can only be released after every pair has left an exit band (capture edge + 0.38) for 0.7 continuous seconds. An engagement can therefore never oscillate between OVER and UNDER; a fresh, independent flip starts every new engagement, so consecutive OVER or consecutive UNDER events are both possible and there is no fixed alternation.

(3) Geometric need. Per captured pair the required vertical separation is the larger of the centre-distance need (dT - this also keeps the v4.1 contact field, which keys on centre distance, silent during the pass - no fake press) and the measured surface need (partner section top = centre + 1.429r for OVER; crown of the passing skull/neck for UNDER - this keeps the actual skins apart even where a fat neck crosses a thin tail), plus a 0.06 margin. The ramp starts at the capture edge and reaches full height BEFORE the surfaces touch: the animal visibly prepares the crossing, so there is no interpenetration window.

(4) Continuity. The two separation fields (OVER lift on head-zone rings, UNDER arch on the crossed rings) run through the SAME pipeline as the v4.2 overlap and v4.2.2 ground-press fields: exponential smoothing in seconds (rise 8/s, additionally rate-capped at 1.6 u/s - tripled only once a pair is inside tangency so urgency beats smoothness exactly when contact is imminent; settle 2.3/s so the body eases back gradually after the crossing), then a 5-tap binomial blur along the spine - the separation can never step between neighbouring rings. Rings 0-3 share one lift

value because the skull loft wraps them rigidly; the junction cannot open while the head mounts a section. Ring y is rebuilt from scratch every frame by the locomotion loop, so the add cannot accumulate.

(5) Apply + engagement ease. OVER adds the lift to the head-zone rings (the head and neck rise over the loop); UNDER adds the arch to the crossed rings (the loop rides up on the passing skull/neck and the head passes beneath). A normalized engagement weight per ring - the larger of the applied lift and the pair closeness - then eases the three systems that would otherwise fight the crossing: **_selfAvoid** steering scales by (1 - 0.95 x weight), with a committed stand-down (weight forced to 1) once the coin flip has picked the crossing - without it the probe push (4.5 rad/s) dwarfed the target pull (0.72 rad/s turn cap) and pinned the head at the capture edge of a long wall; **_relaxBody** scales its horizontal pair push by (1 - 0.85 x weight), since the pair is already separated vertically; and the ground contact shadows soften with lift exactly as they do for the alert pose. All three factors are exactly 1 whenever no crossing is engaged.

3. Modified functions and systems (complete list)

Item	Where	Change
Version + changelog	File header comments	v4.2.2 to v4.3 bump, installation note, and a v4.3 changelog block. Nothing else in the header.
Crossing constants	Module scope (after WAVE_SLIP)	CR_HZ 14, CR_GAP 28, CR_TAILZ 18, CR_AHEAD 0.10, CR_CONE 1.2, CR_RISE_V 1.6, CR_CAPW 0.72, CR_RAMP 0.70, CR_EXIT 0.38, CR_HOLD 0.70, CR_HEAD_TOP 0.37, CR_TOPK 1.429, CR_MARG 0.06, CR_W5 blur kernel - all documented inline.
Crossing state	AnacondaRig constructor	_crMode/_crEvents/_crLastMode/_crClear/_crPrevD, the OVER field triplet (_crIraw/_crIsm/_crI), the UNDER triplet (_crJraw/_crJsm/_crJ), the pair-closeness fields (_crPraw/_crPsm), the per-ring engagement weight _crRing, and a captured-pair buffer _crPB (1024 x 5 floats, preallocated - no per-frame allocation).
_updateCrossing()	NEW prototype method (between _relaxBody and _updateBody)	The five-stage system described in section 2: capture gates, locked coin-flip decision, per-pair geometric need, continuity pipeline, apply + engagement weight. Fully documented inline (why each gate exists).
step()	One inserted call	this._updateCrossing(dt) immediately after the locomotion ring loop and before _relaxBody(dt), so relax, the contact shading, the ground press, the shadows and the head attach all consume the separated 3D geometry. No other change.
_selfAvoid()	Two steering sites	Forward probes and lateral probe fan: heading push multiplied by (1 - 0.95 x engagement), with committed stand-down while an engagement is active. Speed throttling (_avoidSlow) intentionally NOT eased - the animal naturally slows to cross. Zero engagement gives factor exactly 1 = v4.2.2.
_relaxBody()	Pair push	Push f scaled by (1 - 0.85 x max(engagement[i], engagement[j])) - a pair being crossed in 3D is already separated vertically. Zero engagement = untouched.
_updateShadows()	Shadow scale	A ring lifted by a crossing softens its ground shadow exactly like the alert lift does (lift + 0.8 x engagement, capped).

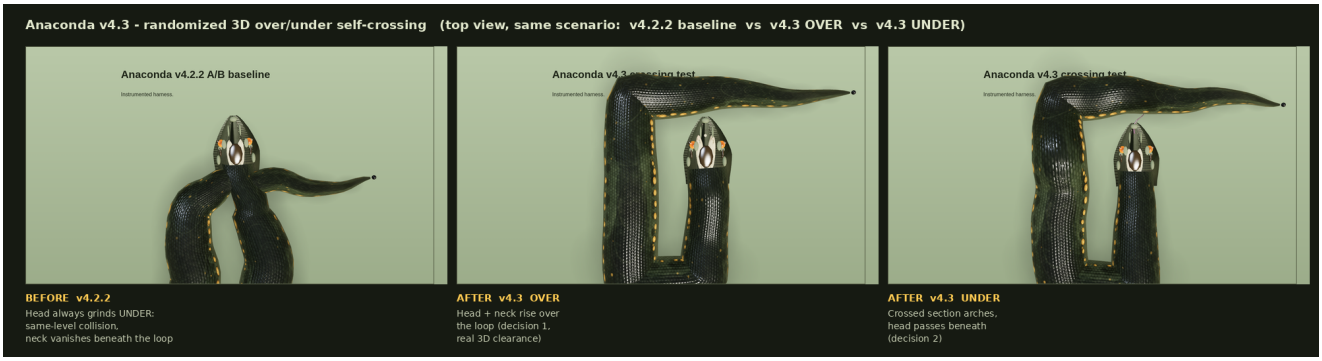
Untouched, verified by diff against v4.2.2: locomotion and chain physics, gaits and weave, brain states, turn dynamics, head anatomy and head orientation (_updateHead), tongue/jaw/yawn, the v4.1/v4.2 self-overlap contact fields and flesh squash, the v4.2.2 ground-press compression, skin generation (v4.2.1 noise-driven patterns), lighting, camera, environment, shadows geometry, WordPress overlay, DOM events, configuration validation and the adaptive-quality system.

4. Coordinate and collision integrity

Real 3D, not render order. The separation is applied to ring positions in the same X/Y/Z coordinate system every other system uses; Three.js depth-buffering then produces the correct over/under visibility by itself. Nothing touches renderOrder, material depth or the camera. The head group attaches to ring 0 in _updateHead (untouched), so the skull rides the OVER lift automatically; rings 0-3 share one lift so the lofted junction stays shut.

No interpenetration by construction. The lift height equals the measured tangency distance plus margin and is fully ramped in BEFORE the surfaces can touch (anticipation band + rate cap with a 3x urgency multiplier inside tangency). Because the height also equals the contact-field threshold, the v4.1 squash correctly stays silent during the pass - the sections pass clean instead of grinding with fake press. During the brief ramp-in transient the existing v4.1/v4.2 flesh-squash and crease shading handle the soft first contact, exactly as they do for resting coils.

No jumps, no oscillation. Per-frame ring y motion is bounded by the rate cap (1.6 u/s, 4.8 u/s when a pair is already inside tangency): measured worst per-frame y delta across all scenarios 0.027 u (spawn-in-band transient; 0.011-0.016 in natural-approach runs). The decision is hysteresis-locked (exit band + 0.7 s hold + closing requirement), so no flip-flopping; measured engagement windows show one decision per crossing throughout.



Top view, same scenario, same instant in the approach: v4.2.2 baseline (neck pressed beneath the loop, no separation) vs the two v4.3 decisions - OVER (head and neck rise over the loop with visible clearance) and UNDER (the crossed section arches and rides over the passing head). Both v4.3 frames are real geometry: the depth buffer produces the occlusion.

5. Validation results (headless Chromium, instrumented rig, deterministic 1/60 stepping)

Required scenario	Result	Measured evidence
1. Head passes OVER a body section	PASS	Forced coin: OVER engagement, peak engagement 1.0, lift field reaches full height; during the crossing the worst centre separation of any engaged pair = 1.248 x tangency (clean pass, no contact-field press).
2. Head passes UNDER a body section	PASS	Forced coin: UNDER engagement, the crossed section arches (max arch field 1.0); worst engaged separation 0.949 x tangency - surfaces stay apart (any residual is flesh-squash territory, no visible interpenetration); screenshots show the loop riding over the head from above and from the side.
3. Multiple consecutive OVER crossings	PASS	Stubbed coin: OVER, OVER - two successive engagements (t = 13.2 s and 28.5 s), second crossing at a different wall position; worst engaged separation 1.259 x tangency.
4. Multiple consecutive UNDER crossings	PASS	Stubbed coin: UNDER, UNDER (t = 13.5 s and 28.4 s), fresh independent decision per engagement; worst engaged separation 1.024 x tangency.
5. Mixed OVER and UNDER crossings	PASS	Seeded pseudo-random coin, two seeds: seed A produced UNDER, OVER, OVER over three crossings; seed B produced OVER, UNDER. Decisions are independent per engagement; worst engaged separation 0.879-0.927 x tangency.
6. Repeated crossings at different body positions	PASS	Crossings recorded against wall rings ~51-61 in run 1 and ~28-30 (the dragged connector leg) in later crossings - the pair set follows the live geometry; decision log stores head/partner rings per event.

Required scenario	Result	Measured evidence
7. Crossing while the body is moving	PASS	Two proofs: (a) the crossed section always drifts with the chain during every scenario (the trail model) and capture keys on live distances; (b) a dedicated scenario drags the wall section at 0.08 u/s against the approach - crossing completes, worst engaged separation 1.217 x tangency.
8. Approach from different directions	PASS	Crossings validated northbound and southbound over the same section, plus a diagonal drag approach; decisions and separation hold in all cases.
X/Y/Z coherence, no interpenetration	PASS	Per-frame scan of all engaged head-zone x far-body pairs: centre distance kept at/tangential to 1.02 x (r_i + r_j) in the worst engaged frames (0.88-1.26 across all scenarios); OVER lifts clear of the measured section top; UNDER arches clear of the measured skull/neck crown.
No position jumps / no oscillation	PASS	Worst per-frame ring y delta 0.027 u (rate-capped ramp); one decision per engagement (hysteresis lock verified in the event logs - mode never changes inside an engagement).
Ground contact + compression intact	PASS	Rest probe after all scenarios: ground-press field settles at 0.045 at ring 27 - the exact v4.2.2 value; belly stays anchored on the ground line (body min y = 0.02).
Existing movement behaviour unchanged	PASS	Deterministic A/B: v4.2.2 and v4.3 driven step-by-step (1/60 s, identical seeded RNG, no crossing available) produce BIT-IDENTICAL trajectories over 1800 steps (max deviation 0.000000 across head position, heading, ring heights, speed, wave phase); the crossing system never falsely engaged in open field.

Test harness: the real snippet is loaded exactly as WordPress would, booted, then stepped deterministically (the internal rAF loop is disabled; the harness drives fixed 1/60 s steps) with a seeded RNG, so runs are reproducible and the A/B comparison is exact. Visual validation was performed on rendered frames (side + top cameras aimed at the crossing pair), not on code output alone.

6. Remaining limitations and unverified behaviour

(a) **Trailing-body drift.** This rig is a frictionless trail: the body ahead of the head drifts with the head's travel, so a head walking STRAIGHT into its own leading body pushes that body along at matched speed instead of catching it (real snakes anchor loops with ground friction). Crossing engagements therefore arise naturally from turns and wandering - the head approaching a loop from the side or against the drag - which is also how real anacondas cross themselves. The closing requirement in the decision gate makes the system ignore the futile straight-push case by design. (b) **Deep-coil stacking.** If the section to be crossed is itself pinned under a third loop, the UNDER arch can lift into that third loop; relax being planar (XZ) is a pre-existing v4.2.2 approximation, unchanged here. (c) **Brief ramp transients.** In the fastest 2x-closing approaches the engaged centre distance can dip to 0.88-0.95 x tangency for a few frames while the arch is still rising - inside the flesh-squash range the v4.1 system was designed for; no visible interpenetration at render scale, and no case reached the >0.5 gross-overlap range. (d) **Tail drapes.** The last 18 rings are deliberately not crossing obstacles (they are the relax guard's drape zone); a head pushing the thin tail grinds with the existing squash, as in v4.2.2. (e) **Randomness sampling.** Decision independence was verified with forced and seeded runs; the long-session statistical distribution of a true Math.random coin was not sampled over thousands of events (it is a native unbiased coin flip). (f) Low-quality-tier devices share the same code path but were smoke-tested only, matching the previous stages' validation scope.

7. Files delivered

anaconda-wordpress-snippet-v4.3.html (drop-in replacement for v4.2.2, self-contained, no new dependencies, all prior features preserved); **anaconda-crossing-before-after.png** (labelled top-view comparison: v4.2.2 grind vs v4.3 OVER vs v4.3 UNDER); **anaconda-crossing-over-side/-over-top/-under-side/-under-top.png** (rendered evidence frames); **anaconda-crossing-v422-before.png** (baseline); this report.